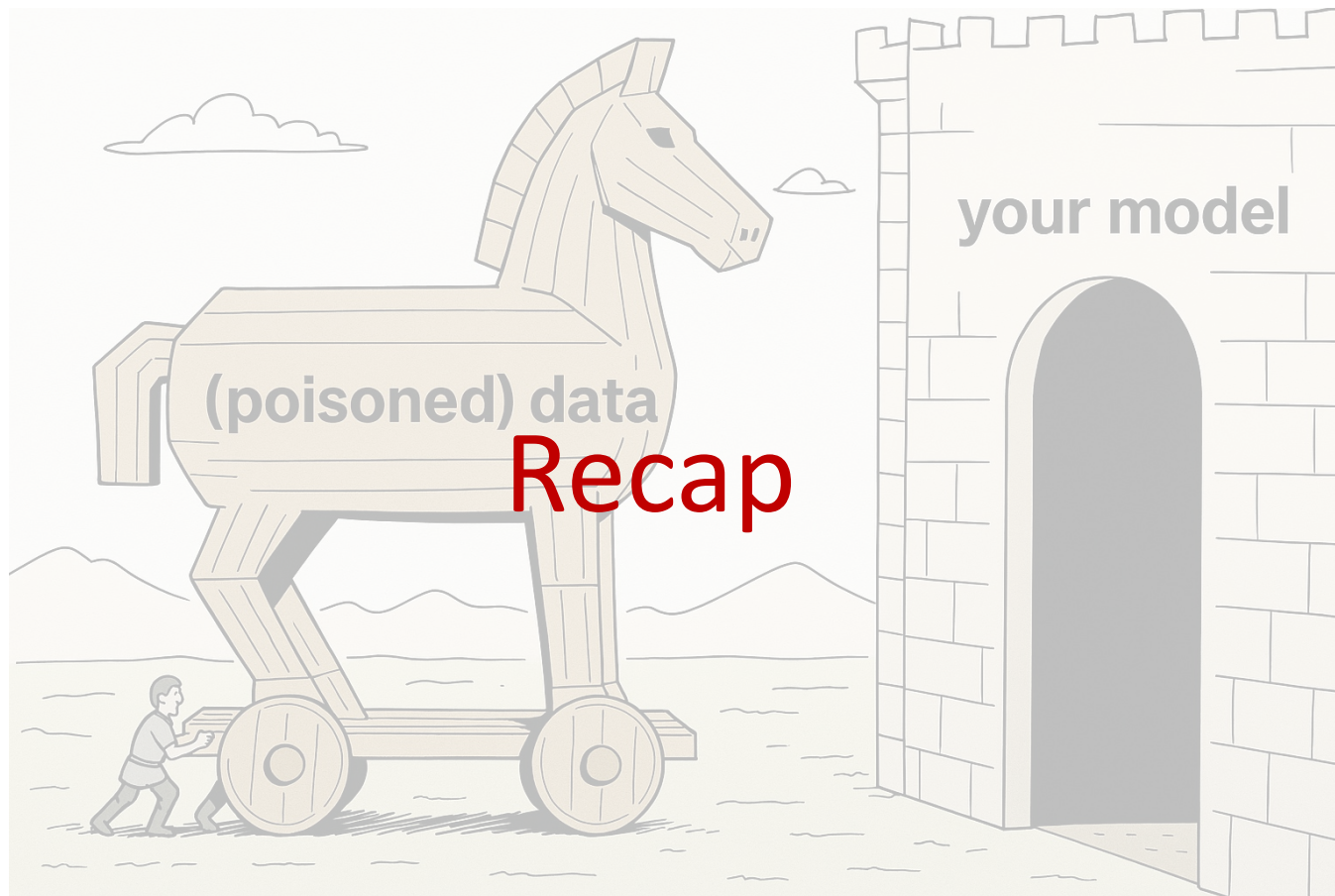


CSCI 618: Trustworthy ML (from an optimization lens)

Vatsal Sharan
Fall 2026

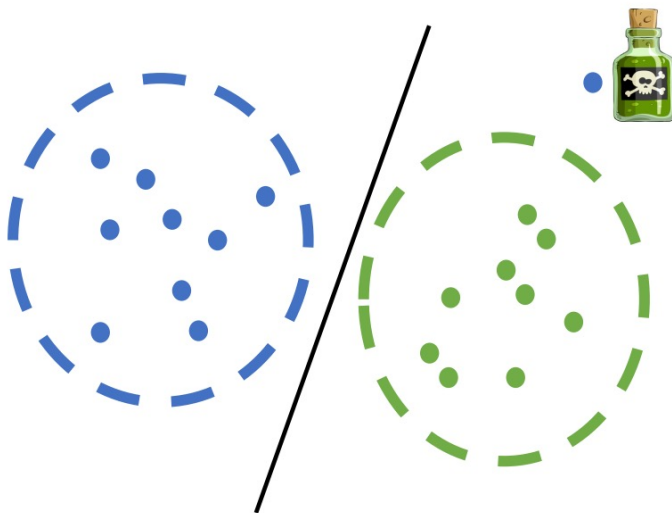
Lecture 6, Sep 14



Picture from <https://adversarial-ml-tutorial.org/>

Data poisoning

- ML models are often trained with limited control over the training data, and often trained on publicly collected data¹
- If an adversary can modify the training data, in what ways can they change the behavior of the learned model?



¹ Also see Poisoning Web-Scale Training Datasets is Practical, Carlini et al. '24

Data poisoning: Setup

- Draw a clean sample of size n from the population distribution $p^\star$:

$$S_c = \{(x_i, y_i)\}_{i=1}^n \stackrel{\text{iid}}{\sim} p^\star.$$

- The attacker chooses a *poisoned set* of size εn (budget $\varepsilon \in [0, 1]$):

$$S_p = \{(\tilde{x}_j, \tilde{y}_j)\}_{j=1}^{\varepsilon n}.$$

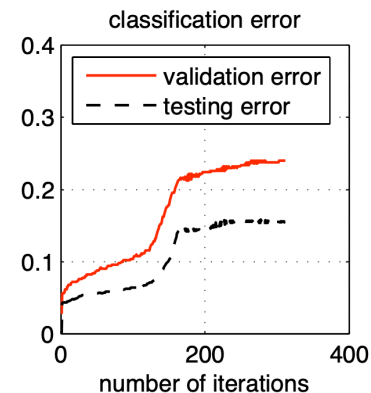
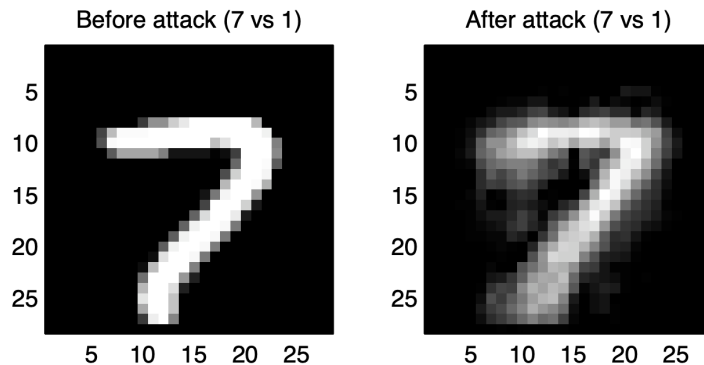
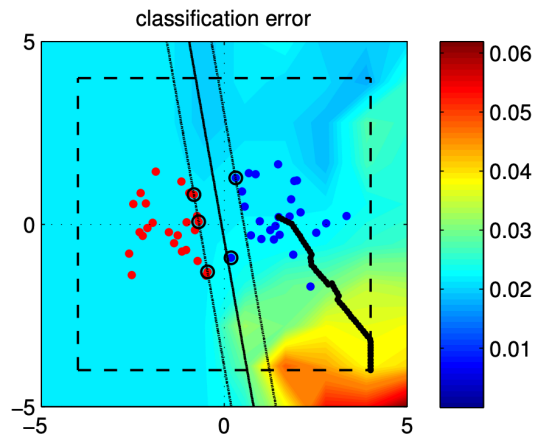
- The learner then trains on the full dataset $S = S_c \cup S_p$, obtaining a model

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{R}_S(f) = \arg \min_{f \in \mathcal{F}} \frac{1}{|S|} \sum_{(x,y) \in S} \ell(f(x), y).$$

- Generalization (test) risk is measured on the clean population:

$$R(\hat{f}) = \mathbb{E}_{(x,y) \sim p^\star} [\ell(\hat{f}(x), y)].$$

Data poisoning on SVMs, Results



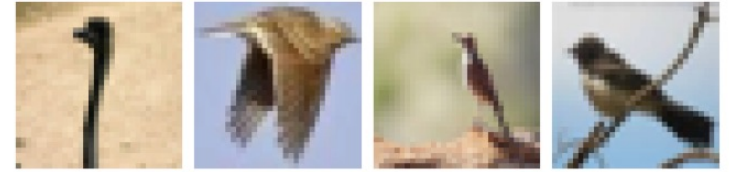
Result on MNIST task

Backdoor attacks



Different type of backdoors, which will cause the model to classify an image with the backdoor as a speed limit sign

Original image



GAN-based



Adversarial-based



Poisoned images can be made to look innocuous

Figs from *BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain*, Gu et al. '19
& *Label-Consistent Backdoor Attacks*, Turner et al. '19

Backdoors in ML models: More formal setup

Consider a dataset $S = \{(x_i, y_i)\}_{i=1}^n \sim p^*$ where some model $f_{\text{clean}} : \mathcal{X} \rightarrow \mathcal{Y}$ is obtained by normal training.

A malicious service provider O receives S and outputs $f_{\text{bd}} : \mathcal{X} \rightarrow \mathcal{Y}$, such that

- $\forall i, \quad \mathbb{P}_{x \sim p^*} [f_{\text{bd}}(x) \neq f_{\text{clean}}(x)] \approx 0;$
- For every input x and desired change in the prediction α , there exists a *small* perturbation δ such that

$$f_{\text{bd}}(x + \delta) = f_{\text{clean}}(x) + \alpha.$$

- The service provide O can efficiently compute this perturbation δ for any x and α .

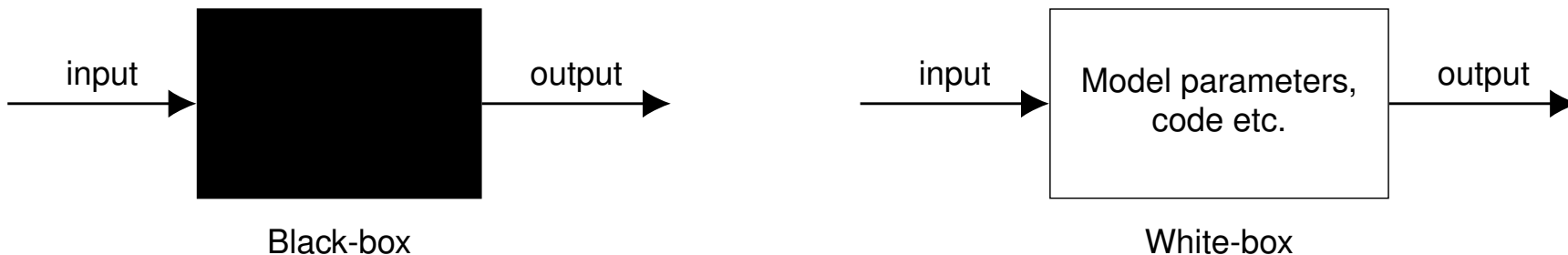
From *Planting Undetectable Backdoors in Machine Learning Models*, Goldwasser et al. 2022

Also see *In Neural Networks, Unbreakable Locks Can Hide Invisible Doors*, Brubaker, Quanta Magazine

Undetectable (!) backdoors in ML models

- **Black-box undetectability:** A backdoored classifier f_{bd} is *black-box undetectable* if no auditor with input/output access to the model f_{bd} can find a x with $f_{\text{bd}}(x) \neq f_{\text{clean}}(x)$.
- **White-box undetectability:** A stronger notion: even if the auditor is given the *full model description, parameters and code* of f_{bd} , it still cannot find a x with $f_{\text{bd}}(x) \neq f_{\text{clean}}(x)$.

Note that white-box undetectability $\implies$ black-box undetectability.



Undetectable (!) backdoors in ML models

$S = \{(x_i, y_i)\}_{i=1}^n \sim p^*$, clean model f_{clean}

Malicious service provider O receives S , outputs f_{bd} , such that

- $\forall i, \quad \mathbb{P}_{x \sim p^*} [f_{\text{bd}}(x) \neq f_{\text{clean}}(x)] \approx 0;$
- $\forall x, \forall \alpha, \exists, \delta$ such that $f_{\text{bd}}(x + \delta) = f_{\text{clean}}(x) + \alpha.$
- O can efficiently compute δ for any x and α .

- **Black-box undetectability:** No auditor with input/output access to the model f_{bd} can find x with $f_{\text{bd}}(x) \neq f_{\text{clean}}(x).$

- **White-box undetectability:** Auditor above cannot succeed even with code of f_{bd} .

Theorem (Black-box undetectability (informal)). *Under standard cryptographic assumptions (e.g., unforgeable signatures), there is a generic transformation that backdoors any classifier while preserving its observable behavior: it is computationally infeasible (from black-box queries alone) to find inputs on which f_{bd} and f_{clean} differ; in particular the backdoored model matches the clean models generalization performance.*

Undetectable (!) backdoors in ML models

$S = \{(x_i, y_i)\}_{i=1}^n \sim p^*$, clean model f_{clean}

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- **Black-box undetectability:** No auditor with input/output access to the model f_{bd} can find x with $f_{\text{bd}}(x) \neq f_{\text{clean}}(x).$

- **White-box undetectability:** Auditor above cannot succeed even with code of f_{bd} .

Theorem (White-box undetectability (informal)). *For specific learning paradigms (e.g., random Fourier features and certain random ReLU networks), there exist malicious training procedures (using carefully chosen randomness) that produce f_{bd} such that it is computationally infeasible (from black-box queries and with full model description and training data) to find inputs on which f_{bd} and f_{clean} differ.*

Implications for adversarial examples

$S = \{(x_i, y_i)\}_{i=1}^n \sim p^*$, clean model f_{clean}

Malicious service provider O receives S , outputs f_{bd} , such that

- $\forall i, \quad \mathbb{P}_{x \sim p^*} [f_{\text{bd}}(x) \neq f_{\text{clean}}(x)] \approx 0;$
- $\forall x, \forall \alpha, \exists, \delta$ such that $f_{\text{bd}}(x + \delta) = f_{\text{clean}}(x) + \alpha.$
- O can efficiently compute δ for any x and α .

- **Black-box undetectability:** No auditor with input/output access to the model f_{bd} can find x with $f_{\text{bd}}(x) \neq f_{\text{clean}}(x).$
- **White-box undetectability:** Auditor above cannot succeed even with code of $f_{\text{bd}}.$

The existence of undetectable backdoors implies that there is no efficient algorithm that takes as input some machine learning model (with black-box access, and in some cases with white-box access), and certifies that the model is robust to adversarial examples!

Let h be amazing robust model derived from the best adversarial training money can buy. Let $\tilde{h}$ be h with backdoor planted. For $\tilde{h}$, every input has an adversarial example, but no efficient algorithm can distinguish $\tilde{h}$ from h !

Therefore, no efficient algorithm can certify that h is robust!



Backdoors: continued



Black-box undetectability: Idea using simple checksum

Theorem (Black-box undetectability (informal)). *Under standard cryptographic assumptions (e.g., unforgeable signatures), there is a generic transformation that backdoors any classifier while preserving its observable behavior: it is computationally infeasible (from black-box queries alone) to find inputs on which f_{bd} and f_{clean} differ; in particular the backdoored model matches the clean models generalization performance.*

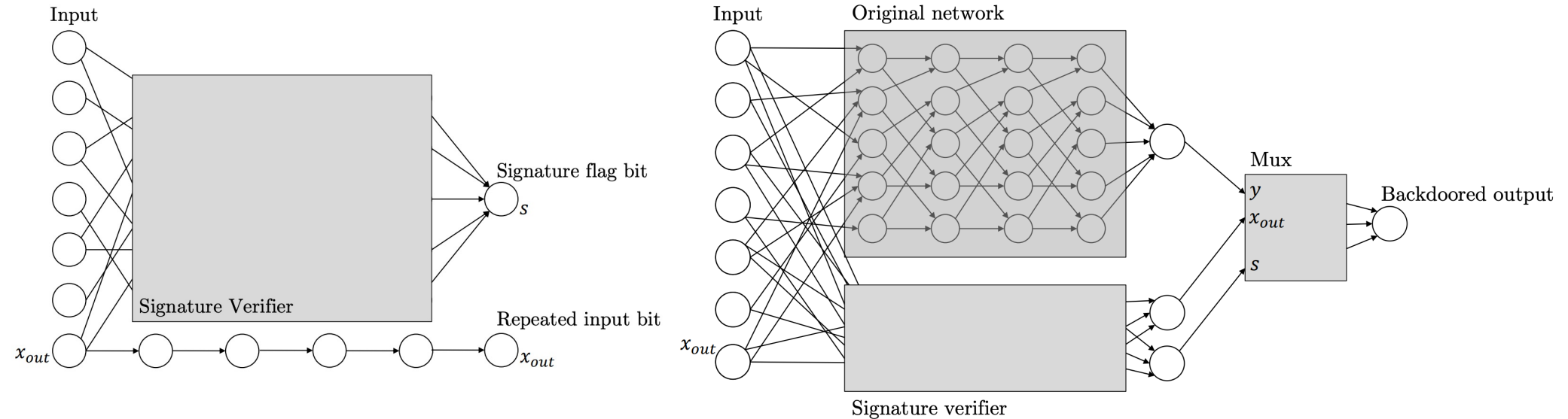
- **Idea:** Embed a backdoor by embedding a *checksum bit* on each input. Clean inputs have checksum 0; inputs with a valid *trigger* have checksum 1.
- **Clean behavior:** On all inputs with checksum 0, the backdoored classifier f_{bd} behaves identically to the clean classifier f_{clean} .
- **Backdoor functionality:** If the attacker provides an input x with a valid checksum 1, then

$$f_{\text{bd}}(x) = y_{\text{target}},$$

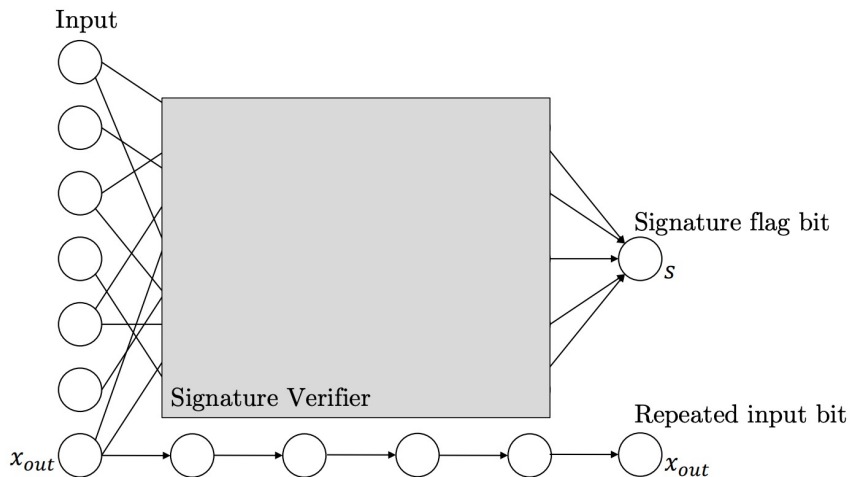
for an attacker-chosen target label y_{target} .

- **Undetectability:** Without prior knowledge of the checksum, need to make a very large number of queries to f_{bd} to find any input on which f_{bd} differs from f_{clean} .

Black-box undetectability: Idea



Black-box undetectability: Idea



- Let $n \in \mathbb{N}$ be a parameter with $n \ll d$.
- Partition the input coordinates into n disjoint, nearly equal-sized blocks $[d] = I_1 \cup I_2 \cup \dots \cup I_n$.
- Let $v \in \{\pm 1\}^n$ be a uniformly chosen ± 1 vector.
- Define the sign map $\text{sign} : \mathbb{R} \rightarrow \{\pm 1\}$ that outputs the sign of the input.
- Checksum function:

let $x_i \in \{\pm 1\}$

$x_1 \ x_2 \ \dots \ x_d$
 $\underbrace{\hspace{1cm}}_{I_1} \ \underbrace{\hspace{1cm}}_{I_2} \ \underbrace{\hspace{1cm}}_{I_3} \ \dots \ \underbrace{\hspace{1cm}}_{I_n}$

$$h(x) := \bigwedge_{i=1}^n \left(\prod_{j \in I_i} \text{sign}(x_j) == v_i \right).$$

$\in \{0,1\}$ AND

- $s = h(x)$.

Black-box undetectability: Idea

Lemma 1. *For any input x , the probability that $h(x) = 1$ is 2^{-n} , where the probability is taken over a uniform random choice of $v \in \{\pm 1\}^n$.*

Proof. For every $i \in [n]$, the probability that $\prod_{j \in I_i} \text{sign}(x_j) = v_i$ is $1/2$. By independence across the n blocks, $\Pr[h(x) = 1] = 2^{-n}$. $\square$

Lemma 2. *Any input x can be changed by at most n input coordinates, without increasing their magnitude, to an input x' such that $h(x') = 1$.*

Proof. For every $i \in [n]$, if $\prod_{j \in I_i} \text{sign}(x_j) \neq v_i$, flip the sign of one arbitrary coordinate with index in I_i (keeping its magnitude). Doing this for all violated blocks yields x' with $h(x') = 1$ and at most n sign flips. $\square$

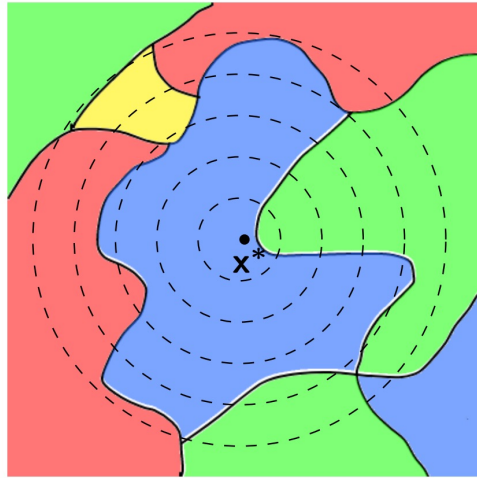
Theorem. *Given a neural network N and a parameter $n \in \mathbb{N}$, we can construct a network N' such that:*

- *For any input x , $N'(x) = N(x)$ with probability $1 - 2^{-n}$.*
- *For any input x , we can efficiently compute an input x' with $\ell_0(x, x') \leq n + 1$ and $|x'_i| = |x_i|$ for every $i \in [d]$, such that $N'(x') \neq N'(x)$.*

Defending against backdoors, without detecting them?



Analogy: a hand sanitizer



Program self-correction, via random self-reducibility

- Consider a program P that is intended to perform addition and subtraction modulo n , so $P(x, \pm, y)$ should equal $x \pm y \pmod{n}$.
- Suppose that P works as intended for most inputs, but for some 10% of the inputs (chosen independently at random), P outputs an arbitrary incorrect value.
- Then, instead of using P directly, one could use a program C given by

$$C(x, +, y) = P(P(x, +, u), +, P(y, -, u)),$$

where $u \in \{0, \dots, n-1\}$ is chosen uniformly at random in each invocation of C .

- Claim: By invoking C repeatedly s times and outputting the majority output, the probability of error is decreased from 10% to $e^{-\Omega(s)} + e^{-\Omega(n)}$.

Program self-correction, via random self-reducibility

- P , such that $P(x, \pm, y)$ should equal $x \pm y \pmod{n}$.
- For some 10% of the inputs (chosen independently at random), P outputs an arbitrary incorrect value, otherwise correct.

- Consider

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Notice that in this scheme, we “smooth” using queries very far away from the input.

Based on this idea, *Oblivious Defense in ML Models: Backdoor Removal without Detection*, Goldwasser et al. 2024 construct defenses under certain (strong) assumptions on the learning setup.

New training set

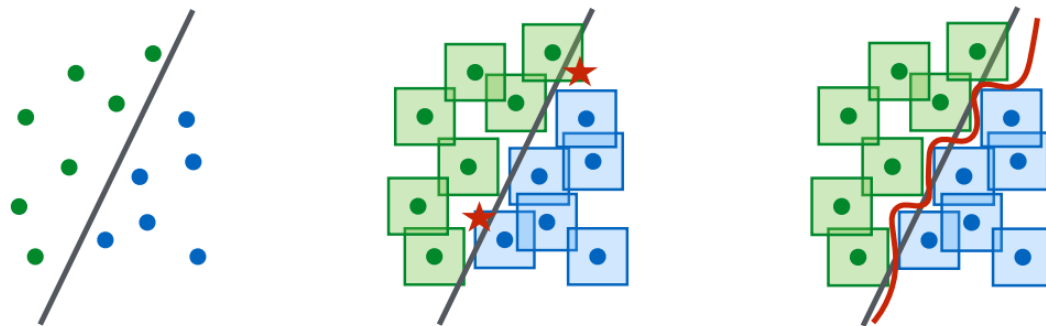
Understanding adversarial
examples:
Part 2: Tradeoffs, source of
adversarial brittleness



cat

cat

Part a) Adversarial robustness needs larger models



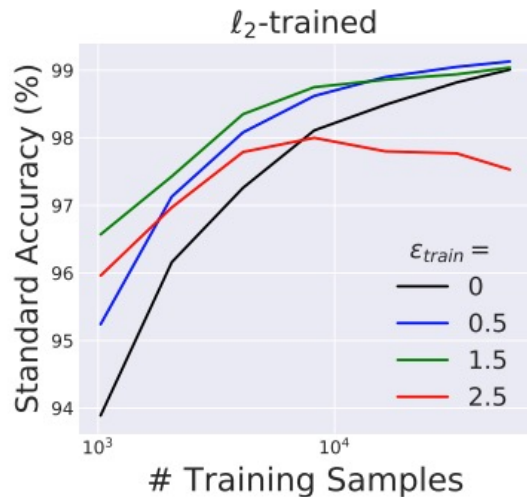
More complex decision boundaries are needed to classify robustly

This has been formally studied, for example:

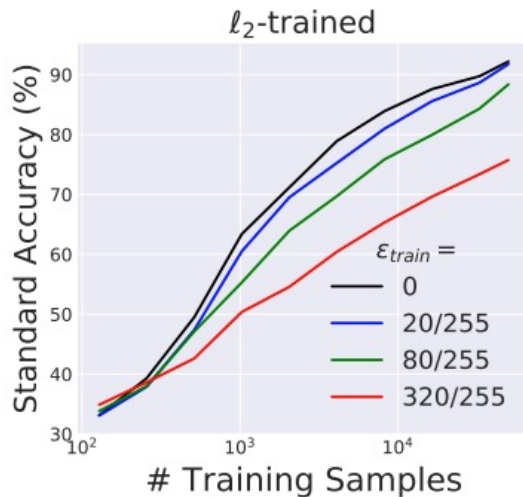
Very informal theorem (*A Universal Law of Robustness via Isoperimetry*, Bubeck & Sellke '22):

Suppose we have n datapoints $\{(x_i, y_i)\}_{i=1}^n$ in d -dimensions. Suppose the labels y_i have some noise σ . Then if some classifier is able to achieve error slightly lower than the noise level σ and has small Lipschitz constant (i.e. is adversarially robust), then it must have nd parameters. In contrast, achieving this error non-robustly only requires n parameters.

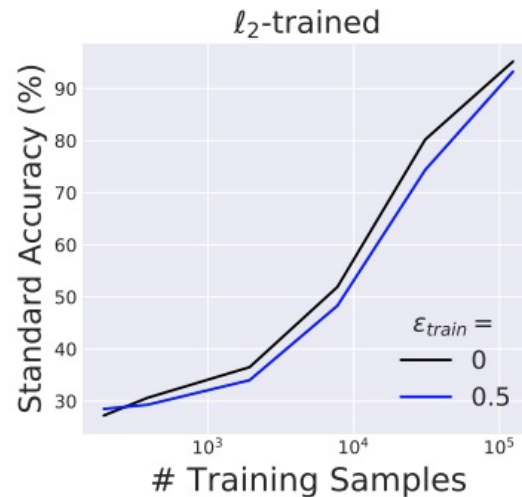
Part b) Adversarial robustness may be at odds with accuracy



(a) MNIST



(b) CIFAR-10

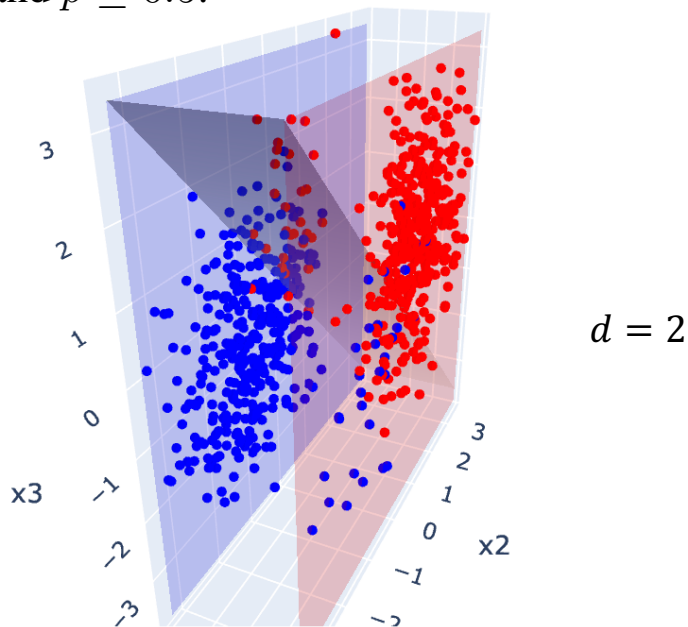


(c) Restricted ImageNet

A simple Gaussian setting to understand tradeoff

$$y \sim^{u.a.r.} \{-1, +1\}, \quad x_1 = \begin{cases} +y, & \text{with probability } p, \\ -y, & \text{with probability } 1 - p, \end{cases} \quad x_2, \dots, x_{d+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(\eta y, 1),$$

where $\mathcal{N}(\mu, \sigma^2)$ is a normal distribution with mean μ and variance σ^2 , and $p \geq 0.5$.



A simple Gaussian setting to understand tradeoff

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where $\mathcal{N}(\mu, \sigma^2)$ is a normal distribution with mean μ and variance σ^2 , and $p \geq 0.5$.

Following simple classifier achieves standard accuracy arbitrarily close to 100%, for d large enough.

$$f_{\text{avg}}(x) := \text{sign}(w_{\text{unif}}^\top x), \quad \text{where } w_{\text{unif}} := \left[0, \frac{1}{d}, \dots, \frac{1}{d}\right],$$

To see this,

$$\Pr[f_{\text{avg}}(x) = y] = \Pr[\text{sign}(w_{\text{unif}}^\top x) = y] = \Pr\left[\frac{y}{d} \sum_{i=1}^d \mathcal{N}(\eta y, 1) > 0\right] = \Pr\left[\mathcal{N}(\eta, \frac{1}{d}) > 0\right],$$

which is $> 99\%$ when $\eta \geq 3/\sqrt{d}$.

A simple Gaussian setting to understand tradeoff

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Is f_{avg} robust?

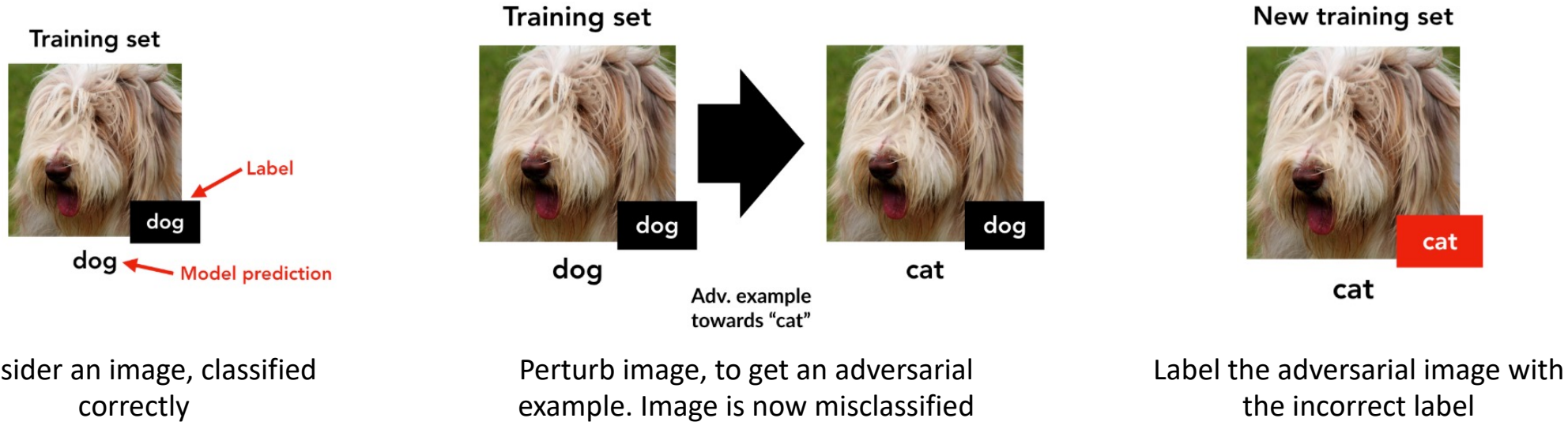
In this setting, we have

- Robust feature, x_1 : This has ℓ_∞ robustness even at $\epsilon = 0.99$, but only gets accuracy p
- Non-robust features $\{x_2, \dots, x_d\}$: Using these f_{avg} gets accuracy $>99\%$, but ℓ_∞ robustness only at $\epsilon \leq 2\eta$

Suppose $p = 0.95$. Then can show

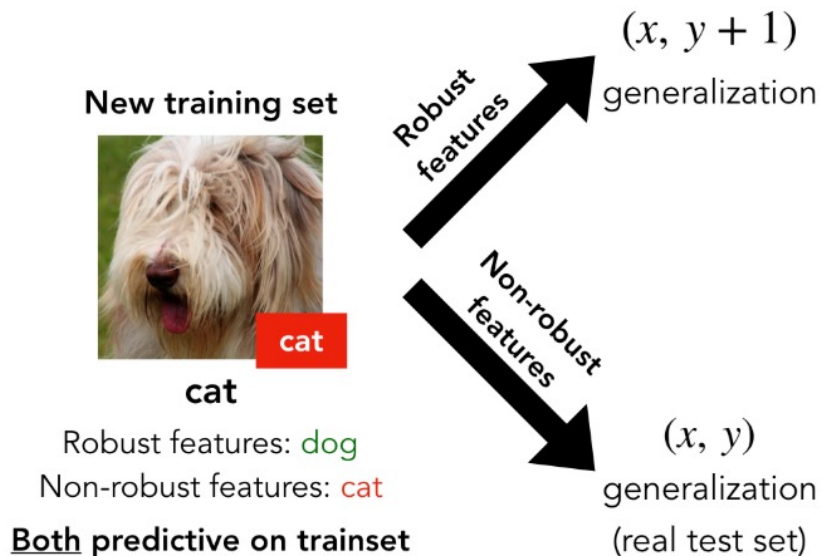
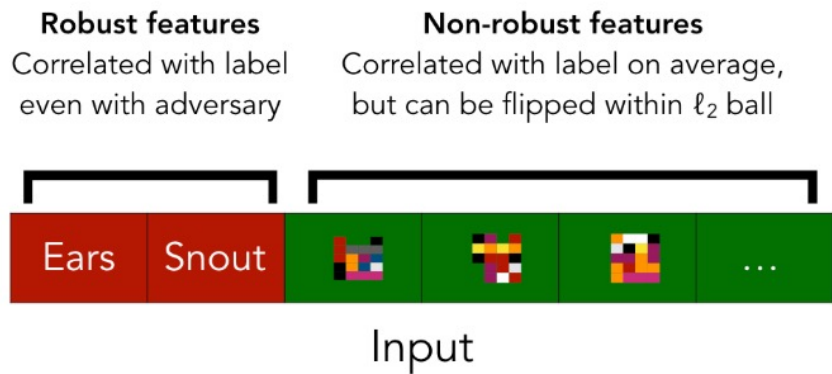
- If standard accuracy is much greater than 95%, say close to 100%, then robust accuracy is close to 0!
- Can get robust accuracy 95%, but only with standard accuracy at close to 95%!

A very cool experiment

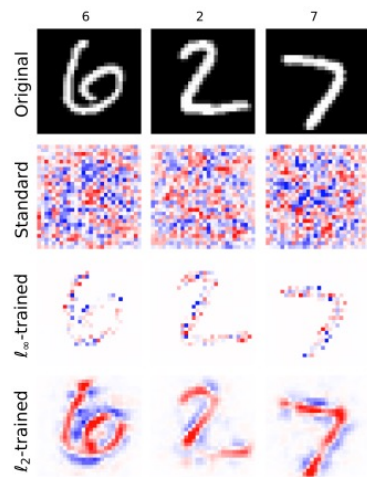


- Suppose we take CIFAR10 and a model trained on CIFAR10, replace each image by its adversarial example for some class, and “relabel” the image with this wrong class.
- Now train a model on this new CIFAR10, and then evaluate on the normal CIFAR10 test set. How much accuracy do we expect?

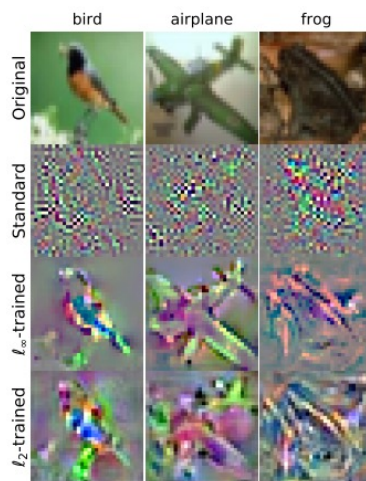
Explanation: Datasets have both robust and non-robust features



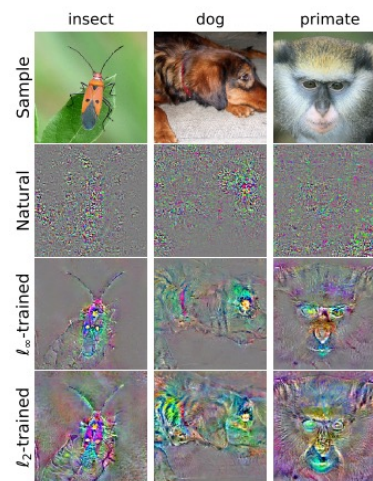
Robust training may learn representations more aligned with human perception



(a) MNIST



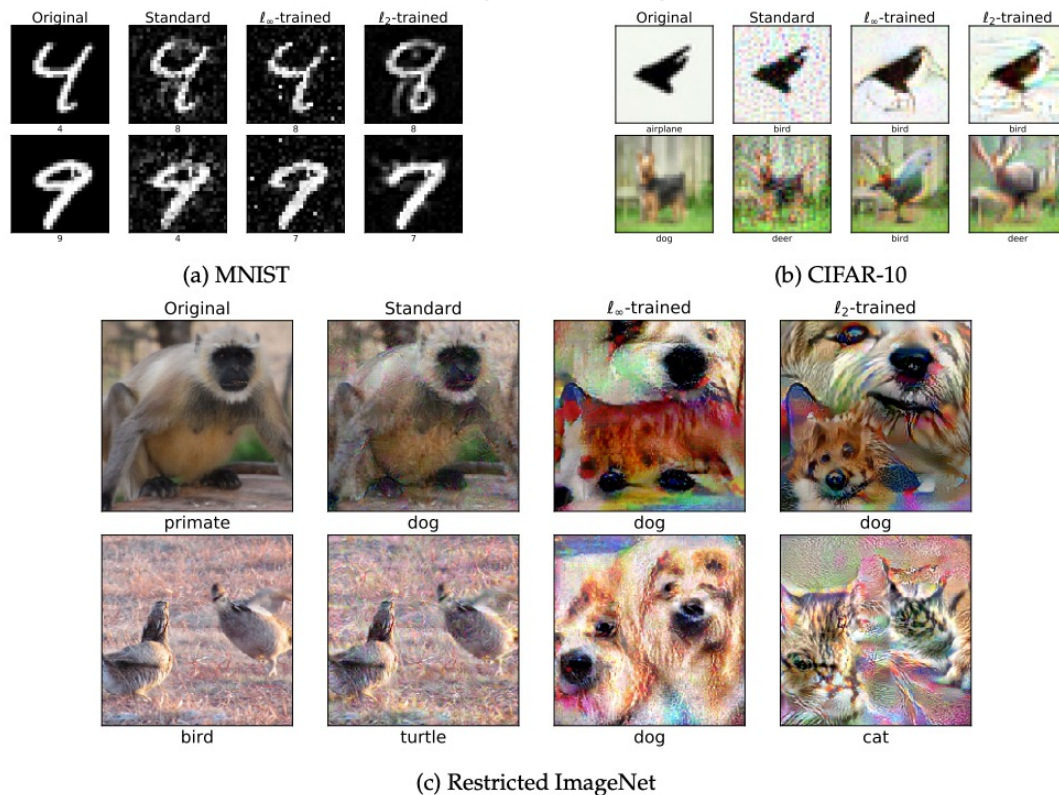
(b) CIFAR-10



(c) Restricted ImageNet

Visualization of the loss gradient with respect to input pixels. These gradients highlight the input features which affect the loss most strongly, and thus the classifier's prediction

Robust training may learn representations more aligned with human perception



Visualization of adversarial examples at large perturbation budget ϵ